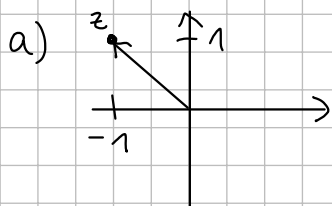
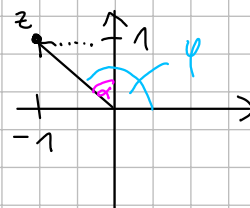


Lösungsvorschlag Probe - Prüfung Lineare Algebra

1. $z = i - 1$.



b) $r = |z| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$
 $\alpha = \arctan\left(\frac{1}{-1}\right) = 45^\circ \hat{=} \frac{\pi}{4}$
 $\varphi = 90^\circ + \alpha = 135^\circ \hat{=} \frac{3\pi}{4}$
 $\Rightarrow \underline{z = \sqrt{2} e^{i \frac{3\pi}{4}}}$

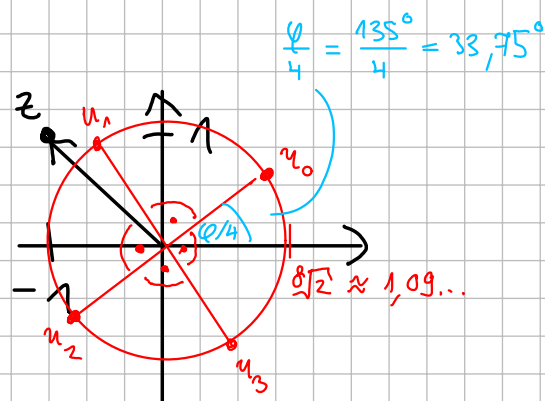


c) $z^{-1} = \frac{1}{z \cdot \bar{z}} \bar{z} = \frac{1}{(-1)^2 + 1^2} (-i - 1) = \frac{1}{2} (-i - 1)$

ODER: $z^{-1} = \frac{1}{\sqrt{2}} e^{i \frac{3\pi}{4}} = \frac{1}{\sqrt{2}} e^{-i \frac{3\pi}{4}}$

d) $u^4 = z = \sqrt{2} e^{i \frac{3\pi}{4}}$ besitzt Lösungen:

$$\begin{aligned} u_0 &= \sqrt[4]{\sqrt{2}} e^{i \frac{3\pi/4}{4}} \\ u_1 &= \sqrt[4]{\sqrt{2}} e^{i \left(\frac{3\pi/4}{4} + 1 \cdot \frac{2\pi}{4} \right)} \\ u_2 &= \sqrt[4]{\sqrt{2}} e^{i \left(\frac{3\pi/4}{4} + 2 \cdot \frac{2\pi}{4} \right)} \\ u_3 &= \sqrt[4]{\sqrt{2}} e^{i \left(\frac{3\pi/4}{4} + 3 \cdot \frac{2\pi}{4} \right)} \end{aligned}$$



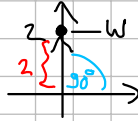
ODER: $u_k = \sqrt[4]{\sqrt{2}} e^{i \left(\frac{3\pi/4}{4} + k \cdot \frac{2\pi}{4} \right)}, \quad k = 0, 1, 2, 3$

2. $w = 2i$

a) $\left(\frac{1}{2}w\right)^{1024} = i^{1024} = \underbrace{(i^2)^{512}}_{-1} = (-1)^{512} = 1.$

b) $w^{-1} = \frac{1}{w \cdot \bar{w}} \cdot \bar{w} = \frac{1}{0^2 + 2^2} (-2i) = -\frac{1}{2}i$

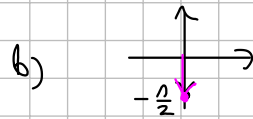
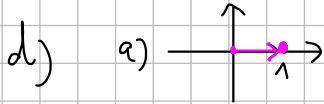
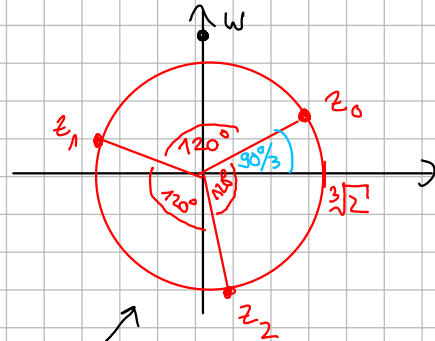
c) $z^3 = w = 2i = 2 e^{i\frac{\pi}{2}}$



$$z_0 = \sqrt[3]{2} e^{i\frac{\pi/2}{3}}$$

$$z_1 = \sqrt[3]{2} e^{i(\frac{\pi/2}{3} + 1 \cdot \frac{2\pi}{3})}$$

$$z_2 = \sqrt[3]{2} e^{i(\frac{\pi/2}{3} + 2 \cdot \frac{2\pi}{3})}$$



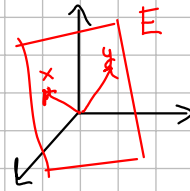
3. $(A|b) = \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 2 & 2a & -1 \end{array} \right) \sim \Pi - 2 \cdot I \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & -2 & 2a-6 & -3 \end{array} \right)$

unabhängig von a immer lösbar & da x_3 frei $\uparrow$ immer ∞ -viele Lösungen,

d.h. $\forall a \in \mathbb{R}$: LGS besitzt ∞ -viele Lösungen.

4.

$$a) \quad E: \lambda x + \mu y = \lambda \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 6 \\ 3 \end{pmatrix}$$



$$b) \quad n = x \times y = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} -2 \\ 6 \\ 3 \end{pmatrix} = \begin{pmatrix} -12 \\ -9 \\ 10 \end{pmatrix}$$

ODER: $n = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix}$ mit $n \cdot x = 0$ & $n \cdot y = 0$

$$n \cdot x = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = n_1 + 2n_2 + 3n_3 \stackrel{!}{=} 0$$

$$n \cdot y = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 6 \\ 3 \end{pmatrix} = -2n_1 + 6n_2 + 3n_3 \stackrel{!}{=} 0$$

$$\text{LGS: } \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ -2 & 6 & 3 & 0 \end{array} \right) \sim \text{II} + 2\text{I} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 10 & 9 & 0 \end{array} \right)$$

$n_3 = \alpha$
frei!

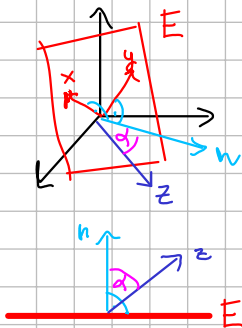
$$\text{II: } 10n_2 + 9\alpha = 0 \Rightarrow n_2 = -\frac{9}{10}\alpha$$

$$\text{I: } n_1 + 2 \cdot \left(-\frac{9}{10}\alpha\right) + 3\alpha = 0 \Rightarrow n_1 = -\frac{12}{10}\alpha$$

z.B. $n_3 = \alpha = 10$ liefert

$$n = \begin{pmatrix} -12 \\ -9 \\ 10 \end{pmatrix}$$

c)



$$\alpha = \angle(z, n) = \arccos \left(\frac{z \cdot n}{|z| \cdot |n|} \right) = \arccos \left(\frac{\begin{pmatrix} -12 \\ -9 \\ 10 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}}{\sqrt{12^2 + 9^2 + 10^2} \cdot \sqrt{1^2 + 2^2 + 3^2}} \right)$$

$$= \arccos \left(\frac{34}{\sqrt{145} \cdot \sqrt{14}} \right) \approx 65,7^\circ$$

$$\angle(z, E) = 90^\circ - 65,7^\circ = 24,3^\circ$$

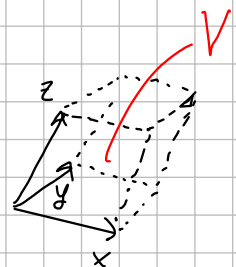
d)

$$V = \left| \det(x|y|z) \right| = \left| \det \begin{pmatrix} 1 & -2 & -1 \\ 2 & 6 & 2 \\ 3 & 3 & 4 \end{pmatrix} \right| =$$

$$= \left| 24 + (-12) + (-6) - (-18) - 6 - (-16) \right|$$

$$= 34$$

(ODER: $V = |(x \times y) \cdot z| = |n \cdot z| = 34$ siehe c))



5. a)

$$x_2 = \frac{\begin{vmatrix} 1 & 1 & 3 \\ 0 & -1 & 2 \\ 0 & 2 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix}} \stackrel{\text{Sarrus}}{=} \frac{-1 - 4}{1} = -5$$

b) EW: $\chi_A(\lambda) = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 2 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^3$, d.h. $\lambda_{1,2,3} = 1$.

EV: $\lambda = 1$: $(A - \lambda E_3) \vec{0} = \left(\begin{array}{ccc|c} 1-1 & 2 & 3 & 0 \\ 0 & 1-1 & 2 & 0 \\ 0 & 0 & 1-1 & 0 \end{array} \right) = \left(\begin{array}{ccc|c} 0 & 2 & 3 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$
 $\uparrow$
 $x_1 = \alpha$ frei

II: $2x_3 = 0 \Rightarrow x_3 = 0$, I: $2x_2 = 0 \Rightarrow x_2 = 0$.

$\Rightarrow \vec{x} = \begin{pmatrix} \alpha \\ 0 \\ 0 \end{pmatrix} = \underline{\underline{\alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}}$



6. $B = \begin{pmatrix} 1 & 2b \\ 3b & 6 \end{pmatrix}$

a) B invertierbar $\Leftrightarrow \det(B) \neq 0$

$$\det(B) = 6 - 6b^2 = 6(1-b^2), \text{ also falls } 1-b^2 \neq 0 \Leftrightarrow \underline{b \neq \pm 1}$$

Dann ist $B^{-1} = \frac{1}{6(1-b^2)} \begin{pmatrix} 6 & -2b \\ -3b & 1 \end{pmatrix}.$

b) B besitzt EW $\Leftrightarrow \chi_B(\lambda)$ besitzt NST.

$$\chi_B(\lambda) = \begin{vmatrix} 1-\lambda & 2b \\ 3b & 6-\lambda \end{vmatrix} = \underbrace{(1-\lambda)(6-\lambda)}_{\lambda^2 - 7\lambda + 6} - 6b^2 = 1\lambda^2 - 7\lambda + (6-6b^2)$$

besitzt NST, falls die Diskriminante $(-7)^2 - 4 \cdot 1 \cdot (6-6b^2) \geq 0$

$$\Leftrightarrow 49 \geq 24(1-b^2)$$

$$\Leftrightarrow \frac{49}{24} \geq 1-b^2$$

$$\Leftrightarrow \underbrace{\frac{49}{24} - 1}_{\frac{25}{24}} \geq -b^2$$

$$\stackrel{(-1)}{\Leftrightarrow} -\frac{25}{24} \leq \underbrace{b^2}_{\leq 0} \quad \forall b \in \mathbb{R} \text{ erfüllt!}$$

d.h. $\forall b \in \mathbb{R}: B$ besitzt EW.

Link von Seite j auf Seite i an Matrixpos. i, j .

7. $A = \begin{pmatrix} L_{11} & L_{12} & L_{13} \\ L_{21} & L_{22} & L_{23} \\ L_{31} & L_{32} & L_{33} \end{pmatrix} = \begin{pmatrix} \frac{0}{n_1} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{n_1} & \frac{0}{2} & \frac{1}{2} \\ \frac{0}{n_1} & \frac{1}{2} & \frac{0}{2} \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 \end{pmatrix}$

↓
↗
↘

anzahl Links auf Seite j

Seite $j = 1 \quad 2 \quad 3$

Seite $j = 1 \quad 2 \quad 3$

Berechne EV zum EW $\lambda = 1$:

$$(A - 1 \cdot E_3 | 0) = \left(\begin{array}{ccc|c} 0-1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 1 & 0-1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0-1 & 0 \end{array} \right) \sim \text{II} + \text{I} \left(\begin{array}{ccc|c} -1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & -1 & 0 \end{array} \right)$$

$$\sim \text{III} + \text{II} \left(\begin{array}{ccc|c} -1 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \begin{array}{l} -x_1 + \frac{1}{2}x_2 + \frac{1}{2}x_3 = 0 \Rightarrow x_1 = \frac{3}{2}x_2 \\ -\frac{1}{2}x_2 + x_3 = 0 \Rightarrow x_3 = \frac{1}{2}x_2 \end{array}$$

↑
 $x_3 = \alpha$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \alpha \begin{pmatrix} \frac{3}{2} \\ 2 \\ 1 \end{pmatrix}. \text{ Für } \alpha = 1 \text{ ergibt sich:}$$

- ①. Seite 2 hat größtes Gewicht $x_2 = 2$
- ②. Seite 1 hat 2. größtes Gewicht $x_1 = 1,5$
- ③. Seite 3 hat 3. größtes Gewicht $x_3 = 1$